

LLM 2-oy 7-Dars Report

Design LLM model - loyihaning umumiy strukturasini tuzish, uy qurishdan oldin chosmasini chosish kabi model qurishdan oldin uni dizaynini yaratish olish samar bo'ladi.

Eng muvofiq beriladigan saroq: qanday model quramiz, nima uchun model quraymiz, Data qayerdan olinadi,

Data Engineering (HTML tag cleaning, PII)

Token / Quality Control / BPE.

PII - shaxsiy ma'lumotga sharoit berish

Configuration

1- ~~8000~~ 8000 das configuration yaml faylda bo'ladi

Configs papkasini ichida run-03.yaml

vocab-size: 8000 - model fashunadigan tokenlar soni

block-size: 256 - foyda-i berishi min block max token

n-embed: 384 - width of the model / vector dimension

n-layers: 6 - model chuqurligi

n-heads: 6 - attention headlar ($384 / 6 = 64$ per head)

dropout: 0.1

ffn-mult: 4 - feed forward is 4x wider than hidden

do-weights: true - output layer reuses the embedding table

Bu tanlangan raqamlar o'zining
foidaligidan, data magduridan va
barcha golga narsalarini o'z ichida
olgan holda bajariladi

① vocab-size

soylariga token ID beriladi - bu da
(0, 1234) gacha. ID berilgan.

soyl $\Rightarrow$ Tokenization (BPE: eng ko'p
foishalanadigan) $\Rightarrow$ Token ID (0, 1234)

Masalan: "Katta o'lma" - [1235, 12, 1236]

OOV (out of vocabulary) - Agar
bu da soylar 1235 ni o'choq ko'r
maka OOV bo'ladi va bunga
muvoziat qilinida Index Error

② block-size yoki context-size

model bor maqtada so'zlar o'ladigan
tokenlar soni

Agar bor juda ko'p soyl berib yabon
sak u holda LLM model bo'lsa

Index Error yoki Unable to
process deb ogohlantirish beradi

③

n-embed - Model bor maqtada

soyladigan jami tokenlar soni

[How word] $\Rightarrow$ Tokenization $\rightarrow$ [Integer ID,

eg. 123] $\rightarrow$ Embedding Lookup $\rightarrow$ [0.5, -0.3, ... 384

floating-point numbers]
- vector dimension & hidden layer

④ n-layer

Bu model chuqurligi, transformer
bloklari soni birladd, birladd
LLM reasoning qobili, complex savolga
javob berishi uchun muhim

Model chuqurligi: Transformer bloklari
soni $\rightarrow$ complex learning

Layer 1 $\Rightarrow$ basic grammar qayta yaratish
sintaksis - kiritiladi

Layer 2 $\Rightarrow$ so'zlarni birlashtirish, qayta
tuzilish

Layer 6 $\Rightarrow$ reasoning, abstract

⑤ n-head

Model birladd 1 tamandan emma
6 tamandan qayta yaratish

$$\text{head dim} = \frac{\text{emb}}{\text{n-head}} = \frac{384}{6} = 64$$

Token
vector

n-emb = 384

64 floats	64 floats	64 floats	64 floats	64 floats	64 floats
Attention Head 1	↓ 2	↓ 3	Attention Head 4	↓ 5	↓ 6

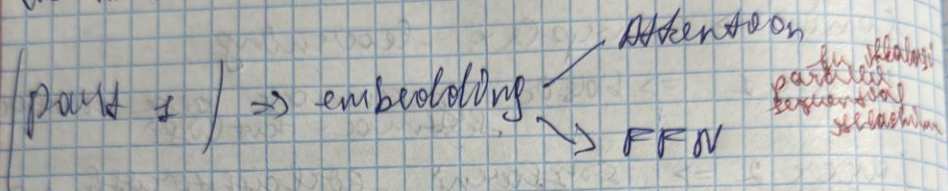
Asosiy qoidalar qoidalar qoidalar
birladd: 11: $384 / 6 = 64$ shu yerda

80% $384 / 4 = 96$ manajer xato birladd

Dropout - o'qitishni oldini oladi
(Memory action) LLM - dropout, LLM - memoy
decay - yamini LLM eavda bu eavda
yengil meshv decay shu yerda

⑦ ffn - feed forward network
 bosqa attention bilan ffn ga
 birlashtiriladi
 Attentionni bosqa bog'lov bilan
 birlashtirish uchun (DB)
 RFN - birlashtirish uchun (DB)

$n_{emb} = 384$ $ffn_mult = 4$ $384 * 4 = 1536$



Misol uchun attention da raqamlar
 ma'lumot har bosqa raqamni
 o'ziga manosa esa birlashtirish
 chiqaradi birlashtirish uchun
 bosqa data base kerak birlashtirish
 FFN bu yerda birlashtirish
 qolishga qardam beradi

⑧ $use_weights = True$

Weightlarni bog'lab qolish
 Input va outputda bir xil ligat
 dan foydalanish

$usb \Rightarrow eng = (olma \Rightarrow apple)$
 $eng \Rightarrow usb = (apple \Rightarrow olma)$

$Seq \Rightarrow Token \Rightarrow TokenID \Rightarrow Embedding \rightarrow FFN$
 $\rightarrow$ Transformer $\rightarrow$ output

Meta data - data haqida ma'lumot

Src/model-config.py - model config data class +
 Scripts /design-model.py - prints the design, checks
 Configs /run-os-game - the actual architecture
 parameters memory
 tokens - per parameter
 numbers